Mahevish Husseni

Data Engineer/Data Analyst

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SUMMARY

- Accomplished Data Engineer with over 5 years of experience in designing and delivering cloud-based data solutions using AWS, Azure, Databricks and Snowflake across healthcare, finance, and enterprise sectors.
- Proficient in architecting and implementing scalable, cloud-native data solutions across AWS, Azure, Databricks, and Snowflake, enhancing data workflows in healthcare, finance, and enterprise sectors.
- Skilled in developing robust ETL pipelines using tools like AWS Glue, Apache Airflow, and Azure Data Factory, improving data accessibility and scalability across distributed systems.
- Experienced with big data technologies such as AWS Redshift, Hadoop, MySQL, and MongoDB, facilitating high-performance analytics and real-time data insights.
- Adept at optimizing OLAP cubes with SSAS, SQL, and DAX, enhancing query performance and enabling efficient data retrieval for cross-functional teams.
- Knowledgeable in programming languages including Python, R, SQL, and Scala for data transformation and analysis, integrating machine learning models to bolster predictive analytics.
- Well-versed in real-time data streaming solutions using Apache Kafka and AWS Kinesis, ensuring immediate data availability for critical decision-making processes.
- Capable of deploying serverless architectures with AWS Lambda and Azure Functions, achieving scalable and cost-effective data processing solutions.
- Experienced in establishing stringent IAM protocols, ensuring compliance with regulations like HIPAA and GDPR, maintaining data security and integrity.
- Skilled in crafting interactive dashboards using Power BI, Looker, and Tableau, empowering stakeholders with real-time insights into revenue forecasting and capital allocation.
- Proficient in automating CI/CD pipelines through tools like Terraform, Jenkins, and AWS CodePipeline, streamlining deployments and enhancing system reliability.
- Committed to upholding data governance standards, adept at handling regulated data to comply with global standards such as GDPR and CCPA.

EDUCATION

International Technological University, San Jose, CA
Master of Computer Science | *Jan 2022 - April 2023*

L.J. Institute of Engineering and Technology, Ahmedabad, India
Bachelor in Information Technology | *Jun 2008 – Jun 2014*

TECHNICAL SKILLS

- **Languages**: SQL, T-SQL, PL/SQL, Python, R, Java, C/C++, Scala, Node.js, JavaScript
- **Databases**: SQL Server, Oracle, MongoDB, DB2, Teradata, BigQuery, PostgreSQL, MySQL, NoSQL, DynamoDB, Redshift, HBase, Presto
- **Big Data Technologies**: Apache Kafka, Apache Airflow, Apache Hadoop, Databricks, Spark, HDFS, Hive, MapReduce, Pig, PySpark
- **Cloud Platforms**: AWS (S3, EC2, Lambda, Redshift, CloudFormation, API Gateway, IAM, Glue, RDS, SageMaker), GCP, Azure (Data Lake, Synapse Analytics, Azure SQL Database, Data Factory, Blob Storage), Snowflake
- **ETL & Data Engineering Tools**: SSIS, SSAS, Informatica PowerCenter, Talend, Snowflake, Data Warehousing, Data Pipelines, Data Governance
- **Reporting & Visualization**: Power BI, SSRS, SAP Business Objects, Tableau, Looker, MS Excel
- **Development Tools**: SQL Query Analyzer, ERWIN, SSMS, SWIFT Tool
- **Libraries & Frameworks**: NumPy, Pandas, Matplotlib, Flask, TensorFlow, Scikit-learn, Keras, Redis
- **DevOps Tools**: Docker, Kubernetes, Terraform, Azure DevOps, Jenkins, Git, GitHub, GitLab, Ansible, CloudWatch, Grafana
- **Methodologies**: Logical and Physical Database Design, UML, Agile, Waterfall, Clustering, Data Mining, Experimental Design
- **Compliance & Security**: GDPR, Role-Based Access Control (RBAC), Encryption

WORK EXPERIENCE

Molina Healthcare, CA
Data Engineer | Jun 2024 – Current

• Contributed to designing data lakes using AWS S3 and Redshift, Databricks, Azure Data Factory (ADF), IICS, DBT Apache Kafka, enhancing data accessibility and improving collaboration across clinical and operational teams.
• Implemented and optimized Spark SQL queries and data processing tasks within the Spark ecosystem, ensuring scalability and performance for processing and analyzing large-scale datasets.
• Assisted in migrating on-premises databases to Amazon RDS, AWS Glue, Databricks, IICS and Informatica, reducing infrastructure costs by 25% and supporting future scalability.
• Played a pivotal role in designing and deploying cloud-based infrastructure solutions for data pipelines, leveraging various AWS services such as CloudFormation, RDS, EC2, Lambda, Athena, S3, DynamoDB, SQS, Redshift, SNS, AWS Glue, CloudWatch, and IAM, with proficiency in Python and PySpark, and incorporating Apache Kafka for real-time streaming.
•Established robust data quality monitoring processes and maintained comprehensive documentation to enhance the user experience, leveraging AWS cloud services, ADF, Kafka, DBT,Informatica, and TIBCO Data Virtualization for both batch and stream processing.
• Implemented, maintained, and managed large-scale data warehousing solutions, ensuring scalability, performance, and reliability.
• Spearheaded the development of an end-to-end data engineering pipeline, from data source identification to schema development, API integration, and data transformation, utilizing AWS services like Amazon S3, RDS, Glue, and GCP services such as BigQuery and Cloud Storage, along with Azure Data Factory and Kafka for hybrid integration.
• Automated ETL pipelines using Python, SQL, and Scala, with Databricks, Informatica,IICS , DBT,Docker, and Kafka on AWS, Azure, and GCP, resulting in a 20% improvement in data accuracy and reduced processing time for healthcare data.

• Incorporated unit testing frameworks such as PyTest and JUnit to validate data pipeline components, ensuring reliability and reducing the risk of downstream data errors.
• Supported the implementation of graph-based analytics with Hadoop, Neptune, and GCP graph tools, providing insights that assisted clinical decision-making.
• Optimized data ingestion processes with AWS Glue, Apache Beam, GCP Dataflow, ADF,DBT and Apache Kafka, reducing manual input errors and streamlining data workflows.
• Collaborated on building CI/CD pipelines with Terraform, Docker, and AWS CodePipeline, ensuring system uptime and smooth updates.
• Assisted in integrating machine learning models into AWS infrastructure via SageMaker, improving predictive analytics for patient care.
• Enhanced data security using RBAC, IAM, and encryption protocols, ensuring compliance with HIPAA and GDP.
•Designed and developed interactive Tableau dashboards to visualize KPIs, track patient outcomes, and provide actionable insights for clinical and operational teams, significantly improving decision-making efficiency.
•Gained hands-on experience in BSA/AML compliance functions including Transaction Monitoring,Customer Risk Rating ,Know Your Customer , and Customer Identification Program, integrating these functions with data pipelines to strengthen risk detection and regulatory reporting.
Analyzed and optimized AWS and GCP billing data using Cost and Usage Reports (CUR) and GCP Billing Export, implementing FinOps best practices for tagging, cost allocation, budgeting, and forecasting to improve cloud spend visibility and cost efficiency.
• Developed cost optimization dashboards in Tableau and Looker, enabling real-time monitoring of cloud expenses and supporting finance teams with data-driven insights for cost control and forecasting.

Blue Cross Blue Shield (BCBS), CA
Data Engineer | Jun 2023 – May 2024

- Developed and automated ETL packages using SSIS, enabling seamless data integration from multiple sources, and improving data reliability.
- Administered and optimized OLAP cubes with SSAS, reducing report generation times by 30%.
- Developed and optimized data processing workflows using Hadoop ecosystem components such as HDFS, MapReduce, and Hive, enabling batch processing and analytics on massive datasets.
- Led the development of end-to-end data pipelines in Azure, incorporating services like Azure Data Factory, Azure Databricks by using Pyspark and Python
- Streamlined data processing by automating ETL workflows in Informatica, Datastage,Matillion and Ab Initio, and Cdata resulting in a 30% reduction in manual handling and improved data processing speed. This initiative enhanced data quality and integrity within Snowflake, with additional integration of Informatica with Airflow for enhanced workflow management.
- Designed, developed, and deployed scalable ETL pipelines in Matillion to integrate data from diverse cloud and on-premises sources into Snowflake, ensuring robust orchestration and automation.
- Implemented Matillion job orchestration with version control and scheduling, enabling auditability and transparency in ETL processes.
- Optimized Matillion transformation components to improve query performance and reduce data processing costs in Snowflake.
- Automated job monitoring and error handling within Matillion, ensuring high availability, improved resilience, and reduced downtime for ETL processes.
- Integrated Matillion with CI/CD pipelines, enabling agile development, faster deployments, and enhanced collaboration across teams.
- Leveraged Matillion's REST API components to ingest third-party APIs and seamlessly integrate external data sources into cloud data warehouses.
- Implemented data quality checks and validations within Matillion ETL workflows, ensuring consistency, accuracy, and trustworthiness of critical datasets.
- Worked on migrating existing ETL jobs to Matillion, modernizing legacy pipelines and reducing infrastructure and maintenance overhead.
- Leveraged AWS Lambda to automate event-driven ETL processes, ensuring near real-time data ingestion and transformation for dynamic reporting and analytics. Lambda was integrated with S3, Glue, and Redshift, providing a scalable, serverless architecture to improve data processing efficiency.
- Developed backend applications to integrate real-time data and automate data loading processes using Oracle SQL, PL/SQL, T-SQL, and Unix Shell scripting. Engaged in SQL performance tuning, troubleshooting, and database administration to ensure optimal database performance.
- Designed, optimized, and secured CI/CD pipelines using Apache Airflow, Docker, Kubernetes, and SQL, collaborating cross-functionally for efficient data processing and analysis
- Configured and managed Apache Kafka clusters to ensure high availability, fault tolerance, and scalability of data streaming infrastructure, handling terabytes of data per day.
- Utilized Snowflake's architecture for external data sharing, supporting data monetization strategies, and collaborated on end-to-end data pipelines, optimizing ETL job performance and modernizing infrastructure, including ETL processes.
- Transitioned legacy reports from SSRS to Power BI, enhancing flexibility and enabling self-service analytics for business users.
- Designed and developed interactive Tableau dashboards to visualize KPIs, improve operational insights, and support executive decision-making across multiple departments.
- Deployed real-time data synchronization using Power BI gateways, reducing latency in reporting and improving data visibility for leadership.
- Collaborated with data science teams to integrate machine learning models into Power BI, improving predictive analytics and decision-making.
- Optimized database query performance by restructuring and indexing, achieving a 20% increase in data retrieval speed.

KPMG, India
Data Engineer | Feb 2019 – Dec 2021

• Developed and optimized ETL processes in Snowflake, leveraging SQL, Snowpark, Informatica, DBT and Snowflake's features to ensure seamless data extraction, transformation, and loading.
• Executed ETL workflows with AWS Glue, Apache Airflow, GCP Dataflow, Informatica,DBT and Apache Kafka, reducing manual intervention and improving the accuracy of financial reporting.
• Designed, implemented, and managed a robust data warehousing solution using Snowflake, GCP BigQuery, DBT and Informatica PowerCenter, enabling the organization to efficiently store and analyze terabytes of data.
• Created new data validation processes using Python, PySpark, and GCP Cloud Data Fusion, along with Informatica Data Quality to ensure data integrity, conducting data profiling and cleansing activities.
• Improved query performance by 30% through in-depth performance optimization, including utilization of Snowflake's advanced SQL features.
• Designed, built, tested, and maintained data management systems using Apache Spark, Apache Hadoop ecosystem tools such as Hive and HDFS, GCP Dataproc, and Apache Kafka, enabling efficient data processing and analytics.
• Implemented real-time data ingestion pipelines using Snowpipe and Apache Kafka, enabling near-real-time access to critical data for analytics and reporting, and worked on disaster recovery and high availability strategies.

• Enhanced data management efficiency and accuracy through proficient utilization of Snowflake features including data retention, Time Travel, clustering, and materialized views, along with GCP BigQuery. Implemented dynamic data masking and fail-safe mechanisms to ensure data integrity and streamline workflows.
• Optimized decision-making processes by leveraging Snowflake functionalities such as data sharing, task scheduling, and dynamic data masking, as well as GCP Looker Studio and Informatica for improved data visualization and orchestration. Enhanced performance of AWS Redshift for high-volume financial data, accelerating data analysis and report generation by 10%.
• Engineered cloud-based data solutions on AWS and Snowflake, simplifying financial model storage and improving system efficiency.
• Developed interactive Power BI dashboards to support real-time financial metrics, driving more informed capital planning and cost management decisions.

Trinity Technolabs, India
Data Analyst | Nov 2017 - Jan 2019

- Designed and automated Tableau dashboards, cutting report generation time by 5% and providing real-time insights for key stakeholders.
- Processed and normalized large datasets using Python and SQL, improving data quality and enabling data-driven business decisions.
- Constructed ETL pipelines with Azure Data Factory, increasing data accessibility and streamlining reporting processes for cross-functional teams.
- Performed comprehensive data cleansing and transformation using Python, pandas, and advanced Excel functions (Power Query, pivot tables, VLOOKUPs), ensuring accurate and consistent reporting outputs.
- Designed automated reporting templates in Excel integrated with Power BI datasets, reducing manual data entry by 40% and enabling real-time business monitoring.
- Utilized SQL to extract and join datasets from relational databases, supporting weekly and monthly business performance reviews.
- Implemented ETL workflows via Azure Data Factory to consolidate data from marketing, finance, and CRM sources into a centralized reporting environment.
- Conducted root-cause and variance analyses for operational inefficiencies, delivering data-backed recommendations that led to process improvements.
- Executed exploratory data analysis (EDA) using Matplotlib and Seaborn to uncover behavioral patterns and business trends in customer and product data.
- Collaborated with cross-functional teams to define KPIs, align reporting needs, and prioritize enhancements to business intelligence assets.
- Maintained version control and ensured collaborative development through Git, contributing to efficient and error-free data pipeline evolution.
- Built regression and classification models using Scikit-learn to predict customer churn and product demand.